# The pigeonhole principle

High school · olympiad thinking · Worked solutions · 02

**Exercise 1.** Show that among 25 people, at least three were born in the same month.

**Solution.** At most two per month would give at most 24 people. Therefore one month contains at least three.

**Exercise 2.** Choose six distinct numbers from $\{1, \dots, 10\}$. Prove that two sum to 11.

**Solution.** The five boxes are $\{1, 10\}$, $\{2, 9\}$, $\{3, 8\}$, $\{4, 7\}$, $\{5, 6\}$. Six choices fill at least one pair.

**Exercise 3.** Prove that among eight integers, two have a difference divisible by 7.

**Solution.** There are seven possible remainders modulo 7. Two integers share a remainder, so their difference is a multiple of 7.

**Exercise 4.** Five points lie in a square of side 2, including its boundary. Show that two are at distance at most $\sqrt{2}$.

**Solution.** Divide the square into four unit squares, assigning each boundary point to just one square. Two points share a unit square, whose diagonal is $\sqrt{2}$.

**Problem · Investigate, justify, conclude.** Choose 11 distinct integers from 1 to 20. Write each as $2^k m$ with m odd and $k \geq 0$. How many values of m are possible? Prove that one selected integer divides another. Show that the conclusion can fail with only ten integers.

**Solution.** Remove all factors of 2. The ten possible odd parts are $1, 3, \dots, 19$. Two chosen numbers share an odd part: $2^a m$ and $2^b m$, with $a < b$ because they are distinct. The second is $2^{b-a}$ times the first. For ten numbers choose $11, 12, \dots, 20$: twice any of them exceeds 20, so none divides another distinct one.